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The Body in the Mind: On the Relationship Between Interoception and Embodiment

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Abstract

The processing, representation, and perception of bodily signals (interoception) plays an important role for human behavior. Theories of embodied cognition hold that higher cognitive processes operate on perceptual symbols and that concept use involves reactivations of the sensory-motor states that occur during experience with the world. Similarly, activation of interoceptive representations and meta-representations of bodily signals supporting interoceptive awareness are profoundly associated with emotional experience and cognitive functions. This article gives an overview over present findings and models on interoception and mechanisms of embodiment and highlights its relevance for disorders that are suggested to represent a translation deficit of bodily states into subjective feelings and self-awareness.

Keywords: Interoception; Interoceptive awareness; Emotions; Time perception; Disturbances of embodiment; Alexithymia; Eating disorder; Self

1. Defining embodiment with respect to bodily and internal signal processing

To a certain extent we all perceive feelings from our body related to the body's internal and external state which provide a sense of our physical and physiological condition. This is related to the basic fact that we “have” a body; that is, that we are “embodied.” These more or less familiar feelings give rise to the intuitive notion that bodily sensations associated with endogenous homeostatic control mechanisms are intrinsically tied to life,

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represent relevant signals for survival and well-being, and underlie mood, emotional state, and fundamental cognitive processes. While it is well known that bodily responding and its perception are key processes in the construction of emotion experience, the fact that bodily processes might be of enormous importance for many more psychological functions, including cognition or decision making, is quite new. Just imagine a very frightening situation when being alone in a dark park and you hear footsteps following you. Nowhere do our mental processes seem to be more clearly embodied than they are in such situations when strong emotions overwhelm us. In this article we try to link bodily processes and their perception to the concepts of embodiment beyond the field of emotions.

2. Interoception and interoceptive awareness: Definitions and measurement

The influence of signals from the bodily interior on experience and behavior has been investigated for more than a 100 years and has profoundly influenced our understanding of our “self.” As a general concept “interoception” includes two forms of perception: proprioception (signals from skin and musculoskeletal apparatus) and viscerosception (signals from the inner organs). A conventional view of interoception held that feelings of visceral and vasomotor activity, hunger, thirst, and other internal sensations were less distinct and less well discriminated and thus different from those that have been associated with the “exteroceptive,” somatosensory system such as pain and itch or temperature. Today, there is evidence that primates have a distinct cortical image of homeostatic afferent activity that reflects all components of the physiological conditions of all tissues of the body. According to the current view, all feelings from the body are represented in a phylogenetically new system that evolved from the afferent limb of the evolutionarily ancient, hierarchical homeostatic system that maintains the integrity of the body (Craig, 2003, 2009b; see 3.1.). This view opens up a wider conceptualization of “interoception,” redefining it from its original narrow usage to refer only to visceral sensation (see Craig, 2008). It comprises sensing the physiological condition of the body, as well as the representation of the internal state within the context of ongoing activities, and is closely associated with motivated action to homeostatically regulate the internal state (Craig, 2008). In this article we will use the terms “interoception” and “interoceptive” according to this concept, with a main focus on the perception and processing of internal bodily, visceral signals.

Research on interoception of the cardiovascular (Critchley, Wiens, Rotshtein, Ohman, & Dolan, 2004; Herbert, Pollatos, & Schandry, 2007a; Pollatos, Schandry, Auer, & Kaufmann, 2007c; Schandry, 1981) and the gastrointestinal system (Stephan et al., 2003) and from pain research (Hosoi et al., 2010) underscores that there are significant interindividual differences in “interoceptive awareness” (IA). Basic research on IA has predominantly focused on heartbeat perception and the individual sensitivity for cardiac signals (“cardiac awareness”). This sensitivity has been most usually quantified by using validated and reliable heartbeat perception tasks (Jones, 1994; Wildmann & Jones, 1982), such as “tracking” (Dale & Anderson, 1978; Schandry, 1981) or “discrimination tasks” (Brener, Liu, & Ring, 1993; Whitehead & Drescher, 1981). In these tasks participants are instructed to perceive

their own heartbeats without feeling for their pulse. They allow calculating individual heart-beat perception scores that characterize the deviation of the subjectively felt cardiac signal from the objective, “true” cardiac signal, that is, the individual heartbeats

The individual degree of IA can be conceptualized as a trait-like sensitivity toward one’s visceral signals. However, the perceptibility of visceral, cardiac signals can also be manipulated by procedures that evoke changes in autonomic cardiovascular activity (Schandry, Bestler, & Montoya, 1993) or the training of concentrating on one’s cardiac activity (Schandry & Weitkunat, 1990). IA as assessed by cardiac awareness is related to greater sensitivity to emotional responding and cardiovascular autonomic reactivity in different situations evoking autonomic changes (Herbert et al., 2012; Herbert, Pollatos, Flor, Enck, & Schandry, 2010b; Pollatos, Herbert, Matthias, & Schandry, 2007b), proposing that cardiac awareness can also be the result of a “visceral” learning process. This is suggested to depend on greater autonomic reactivity during different situations of daily life evoking substantial changes in autonomic activity that is probably associated with greater repetitive activity of relevant “interoceptive” brain regions (Herbert, Herbert, & Pollatos, 2011; Herbert et al., 2010b). Referring to these ideas, the possibility should be taken into consideration that observed interactions might be formed in the opposite direction, with higher IA leading to enlarged autonomic response via top-down modulated attentional processes.

It is not yet clear in how far the interoceptive perception of signals coming from different bodily systems converges into general IA or if there are individual differences across modalities. Both possibilities seem feasible. To date there are only sparse data showing that cardiac awareness correlates with the ability to detect changes of the activity of the stomach (Whitehead & Drescher, 1981). However, Whitehead and Drescher suggested that there may be a generalized tendency to be aware of visceral events, at least in specific situations evoking interoceptive feelings. In the following section we will have a closer look on underlying systems supporting interoception.

3. Interoception as a basis of embodied processes

3.1. Evidence from neuroanatomy

Neuroanatomic evidence emphasizes the relevance of an “interoceptive neural network” in the brain comprising the somatosensory and somatomotor cortices, the insular cortex, cingulate cortex (ACC), and prefrontal cortices (ventromedial prefrontal cortex, dorsolateral prefrontal cortex). These structures are relevant for monitoring the internal emotional and viscerosensory state (Critchley, Corfield, Chandler, Mathias, & Dolan, 2000; Critchley et al., 2003), for emotion processing and reactivity (Phan, Wager, Taylor, & Liberzon, 2002), for the feeling of self-generated and externally induced emotions (Anders et al., 2004), and the self-regulation of feelings and behavior (Beauregard, Levesque, & Bourgouin, 2001; Bechara, 2004).

Within this interoceptive network the insula represents a relevant projection site of viscerosensory input from different modalities from the body. Interoceptive stimuli that activate the

anterior insular cortex (AIC) include thirst, dyspnea, the Valsalva manoeuver, “air hunger,” sensual touch, itch, heartbeat, and distension of the bladder, stomach, or esophagus (see Craig, 2009b). Evidence from work on recognition of body movement, music and rhythm, emotional awareness, self-recognition, and time perception underscores the relevance of the insular cortex and/or the ACC (Craig, 2009b) for *all* subjective human feelings.

Findings on the neural basis of interoception by using heartbeat perception confirm the relevance of the interoceptive neural network and show that good compared to poor heart-beat perceivers demonstrate greater activation, especially in the right AIC (Critchley et al., 2004; Pollatos et al., 2007c). The weight of evidence suggests greater central representation and integration of cardiovascular signals in persons who are more aware of their cardiac signals. A recent model of interoception (Craig, 2009b) underscores the role of the insula for the translation of visceral and further bodily states into subjective feelings and self-awareness. Accumulating evidence (Craig, 2008, 2009a,b) highlights the relevance of a phylogenetically new homeostatic afferent lamina-1 spinothalamocortical pathway that converges to “interoceptive centers” in the insular and orbitofrontal cortices (Craig, 2009a,b) and gives rise to conscious visceral perception (see Fig. 1).

It is suggested that different portions of the insula are involved in different and successive steps of neural processing building the basis of the sequential integration of the primary homeostatic condition of the body with salient features of the sensory environment and with motivational, hedonic, and social conditions. Raw interoceptive signals such as those coming from visceral changes and pain, first project to the posterior insula and become progressively integrated with contextual motivational and hedonic information as they progress toward the anterior insula. The neural constructs of the distinct, individually mapped feelings in the posterior insular cortex are then re-represented in the mid-insula that integrates

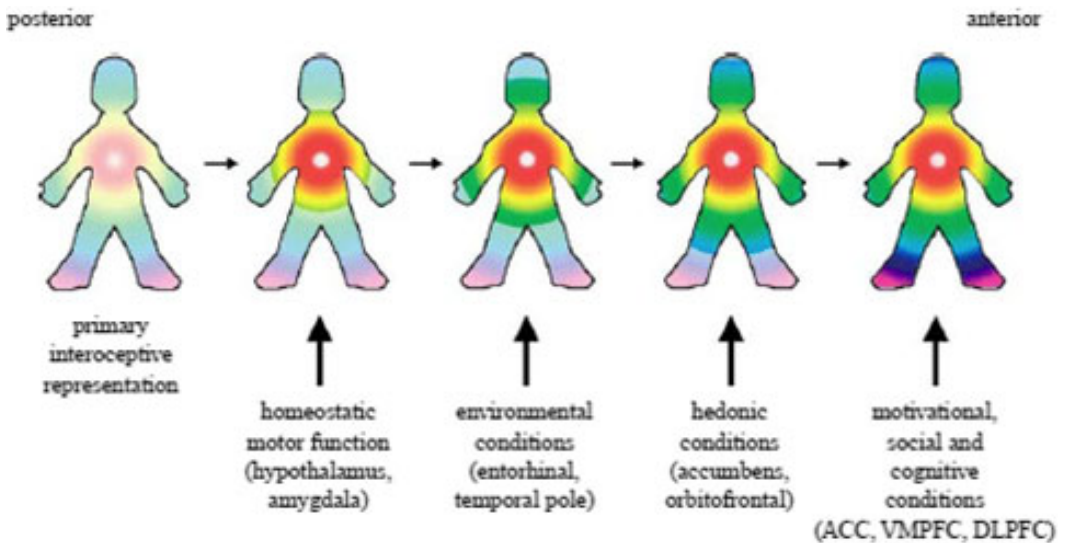


Fig. 1. Suggested posterior-to-anterior progression in the insula (VMPFC, ventromedial prefrontal cortex; DLPFC, dorsolateral prefrontal cortex; figure adapted from Craig, 2009a).

the homeostatic re-representations with activity associated with emotionally salient environmental stimuli of many sensory modalities from different parts of the brain.

The culmination point of this progression is a complex and rich representation of the “global emotional moment” in the anterior insula that represents the ultimate representation of all one’s feelings, thereby constituting the “sentient self,” in the immediate present (now). In this model, the AIC provides a unique neural substrate that instantiates all subjective feelings from the body and feelings of emotion at a certain moment of time supporting the proposition that subjective awareness is built on homeostasis. In this view, the neural basis for “awareness” is the neural representation of the physiological condition of the body, and the homeostatic neural construct for a feeling from the body is the foundation for the encoding of all feelings (Craig, 2009b).

This is in accordance with the “somatic marker” hypothesis of Damasio (1994, 1999) stating that this meta-representation of bodily states constitutes an emotional feeling, accessible to consciousness and providing the “gut-feeling” that guides our decision processes and forms the basis for our “self” and consciousness. The neuroanatomic basis of interoception represents the link for the “body in the mind” and the mechanisms of the embodiment of affective and cognitive functions.

3.2. The role of interoception for decision making, emotions, and behavior: Signs of embodiment

William James (1884) stated that the experience of emotion could be defined as the perception of bodily responses and following models (Craig, 2009b; Damasio, 1994, 1999) suggest that the foundation for our emotional feelings lies in the neural representation of the physiological condition of the body, with “somatic markers” evoking feeling states that influence cognition and behavior. Theories of embodied cognition hold that higher cognitive processes operate on perceptual symbols and that concept use involves reactivations of the sensory-motor states that occur during experience with the world (e.g., Niedenthal, 2007). Similarly, activation of interoceptive representations and meta-representations of bodily signals supporting IA are profoundly associated with emotional experience and cognitive functions. A recent study performed by Tsakiris and colleagues [1] revealed that IA modulated the integration of multi-sensory body-percepts, thus highlighting the important role of interoception and IA for multi-sensory integration.

There is ample evidence suggesting that IA is crucial for the intensity of emotional experience (e.g., Barrett, Quigley, Bliss-Moreau, & Aronson, 2004; Herbert, Herbert, and Pollatos, 2007a, 2010a,b; Pollatos, Gramann, & Schandry, 2007a; Pollatos, Kirsch, & Schandry, 2005; Wiens, 2005) and the higher order processing of emotional stimuli (Herbert et al., 2007a; Pollatos et al., 2005). Studies employing decision-making tasks and risk manipulation provide evidence that deficits in the generation and/or representation and processing of physiological arousal are profoundly associated with disadvantageous and more risky decision behavior (Bechara, 2004; Bechara, Damasio, Damasio, & Anderson, 1994; North & O’Carroll, 2001). Recent data confirm that the accuracy with which bodily, cardiac signals are perceived is associated with benefits in decision making (Werner, Jung,

Duschek, & Schandry, 2009). Moreover, IA has been shown to constitute a decisive factor for the behavioral self-regulation in situations that allow for the self-control of behavior such as physical workload (Herbert, Ulbrich, & Schandry, 2007b). These findings indicate that IA is crucially associated with the self-regulation of behavior in different situations of daily living that are accompanied by somatic and/or physiological changes giving rise to “somatic markers.” The relevance of our “gut-feelings” for decision making and behavior especially shows up in situations of uncertainty and complexity where we are free to decide upon our own actions (Damasio, 1994). Furthermore, the importance of interoceptive brain structures, especially the AIC, has been shown for risk prediction (Preuschoff, Quartz, & Bossaerts, 2008) and feelings of anticipated value during purchase and sales decisions (Knutson, Rick, Wimmer, Prelec, & Loewenstein, 2007). These insights have become a relevant part of research in neuroeconomics (Loewenstein, Rick, & Cohen, 2008).

Further results report that cardiac awareness is positively associated with benefits in selective and divided attention (Matthias et al., 2009), suggesting greater IA to represent an indicator of greater attention allocated toward both internal and external relevant events as well as self-focused attention. These results highlight the “visceral” embodiment of emotional and cognitive processes. Taken together, convincing empirical evidence and theoretical foundations have been provided for the relevance of interoception for feelings and cognitive functions. However, it should be borne in mind that most of the findings up to now are based on correlational data and imply the objection to ask in how far interoception is indeed causally involved in our emotional and cognitive processes.

In order to answer this question, future studies are necessary to manipulate interoceptive signal processing under experimentally controlled conditions and to investigate its effects on emotional and cognitive functions. There is initial evidence showing that the training of heartbeat perception or experimentally evoked changes in autonomic nervous system activity and associated cardiodynamic functions induce an improvement of the perception of internal cardiac signals. This improvement is in turn related to an enhancement of brain functions reflecting cardiac signal processing as well as to an increase in emotional experience (Schandry & Weitkunat, 1990; Schandry et al., 1993).

3.3. Embodiment of time perception

The perception of time is part of human experience; it is essential for everyday behavior and for understanding any kind of complex behavior. Recent debate connects time perception with bodily responses and embodiment by assuming that physiological states and emotions associated with changes in physiological states underlie our perception of time (Craig, 2009b; Wittmann, 2009). Such a direct link between the perception of time and physiological processes has been proposed by Craig (2009a) who claims that our experience of time relates to emotional and visceral processes because they share a common underlying neural system, the insular cortex and the interoceptive system. Wittmann (2009) follows that since emotions and physiological states seem so fundamental to the experience of time, it is tempting to assign a pivotal role to these processes related to a core timekeeping system. In

line with these hypotheses, it is conceivable that the number and rate of body signals accumulated in the insula over a given timespan create our perception of duration.

In a recent study Wittman and co-authors (Wittmann, Simmons, Aron, & Paulus, 2010) found empirical evidence for this assumption: They demonstrated that during the encoding of time intervals (9 and 18 s tone intervals) activation curves over time show an accumulating pattern of neural activity, which peaks at the end of the encoding interval within bilateral posterior insula and superior temporal cortex. They argue that the accumulation function in the posterior insula might be correlated with the encoding of time intervals as suggested by Craig (2009a,b). Craig's model proposes a close interaction between interoceptive processes and time perception, suggesting that our experience of time emerges from emotional and visceral states processed in the insular cortex. It can be interpreted as a series of global emotional moments that are indexed across a finite period of present time, from the past into the anticipated future. These series produce a cinemascopic "image" of the sentient self that is continuous across a moving window of present time. Across any point of such consecutive processing hierarchies problems can occur. In the following we will try to highlight some examples of disturbances in embodiment that are very clearly associated with abnormalities in interoceptive functioning.

4. Disturbances of embodiment: Psychopathologic phenomena as translation difficulty of bodily processes in constituting the self

In a recent article Fuchs and Schlimme (2009) presented the concept of embodiment as a novel paradigm for an interdisciplinary approach to psychopathology and neuroscience. The authors argued that the phenomenology of the lived body is able to overcome dualistic concepts of the mind as an inner realm of representations that mirror the outside world. They referred to a common phenomenological distinction between the body that one pre-reflectively lives in, that is, the lived or subject body (German: *Leib*), and the physical body that one can perceive or that is perceived by others, in other words, the object body (German: *Körper*). Fuchs and Schlimme defined psychopathology of embodiment based on the proposed distinction of subject and object body. Accordingly, disturbances of embodiment may be classified (a) as primarily affecting the subject body or pre-reflective embodied sense of self; as is the case, for example, in schizophrenia or depression, or (b) as being related to the body image or explicit body awareness. The latter includes, for example, body dysmorphic disorder, somatoform disorders, or eating disorders (EDs) such as anorexia nervosa. In the following paragraph we want to refer to the idea of defining psychopathologic processes in terms of affected embodiment and want to introduce two clinical examples, namely alexithymia and EDs. For both entities there is a clear hypothesis that a disturbance in embodiment is related to affected perception of bodily processes. In this context, recent work on body-ownership and its representation in the brain are of great relevance (Tsakiris, Jimenez, & Costantini, 2011; Tsakiris, Schütz-Bosbach, & Gallagher, 2007). Namely, Tsakiris and colleagues (Tsakiris et al., 2011) were the first to demonstrate an interaction between interoceptive and exteroceptive awareness of the body. They were further able to show that

interoceptive sensitivity predicts the malleability of body representations, suggesting that IA modulates the online integration of multi-sensory body-percepts.

4.1. *Alexithymia*

Alexithymia, a syndrome that involves a marked inability to identify, describe, regulate, and express one's emotions (Sifneos, 1976; Taylor & Doody, 1985) has been related to a broad range of physical and psychiatric disorders (e.g., EDs, depression, posttraumatic stress disorders etc.). At the present time, both within clinical and nonclinical populations, alexithymia is considered a continuous personality trait, with persons differing in their ability to identify and describe their feelings. The construct of alexithymia is most widely assessed by self-report questionnaires such as the Toronto Alexithymia Scale (Bagby, Parker, & Taylor, 1994) or the Bermond-Vorst Alexithymia Questionnaire (Vorst & Bermond, 2001), as well as by observer-rated questionnaire (Haviland, Warren, & Riggs, 2000). A performance measure, the Levels of Emotional Awareness Scale (LEAS; Lane, Quinlan, Schwartz, Walker, & Zeilin, 1990), is also well-validated. Despite a very complex picture concerning convergent and divergent validity of different measures of alexithymia (Berthoz, Perdereau, Godart, Corcos, & Haviland, 2007; Subic-Wrana, Bruder, Thomas, Lane, & Kohle, 2005), there is broad consensus that a core problem of alexithymia refers to deficits in emotional awareness (Coffey, Berenbaum, & Kerns, 2003; Lane et al., 1998).

Lane and colleagues (Lane, Ahern, Schwartz, & Kaszniak, 1997) introduced alexithymia as equivalent of emotional blindsight and re-defined this concept recently by using the term "affective agnosia." Affective agnosia is characterized by a present habitually emotional response (viscero- and somato-motor responses) with a lack of either recognizing or experiencing this response pattern as an emotional feeling state. We here would like to introduce the idea that alexithymia as a personality trait is characterized by a disturbance in translating bodily signals to conscious awareness going beyond the field of a classification as an affective disorder. In a recent study we could demonstrate that high scores of alexithymia are associated with low IA (Herbert et al., 2010a) as assessed by a heartbeat perception task. This study adds clear evidence that alexithymia can be described in terms of a disturbance in embodiment. Our results throw a new light on alexithymia and its conceptualization as impairment in the capacity to consciously access emotional responses (Lane et al., 1998; Luminet, Vermeulen, Demaret, Taylor, & Bagby, 2006; Taylor, Bagby, & Parker, 1999) by demonstrating deficient perception of bodily signals to represent a habitual characteristic in alexithymia.

4.2. *Eating disorders*

Eating disorders are the most prevalent psychiatric disorders in females aged between 14 and 26 years and are associated with considerable physical and psychological morbidity. In the last decades, the frequency of these illnesses has greatly increased, representing a great challenge for physicians of various specialties and significantly impacting health care in the female population. Former research (Fassino, Pierò, Gramaglia, & Abbate-Daga, 2004;

Lilenfeld, Wonderlich, Riso, Crosby, & Mitchell, 2006; Matsumoto et al., 2006) reported a decreased ability to discriminate hunger and satiety sensations in EDs as measured by questionnaire (IA score of the eating disorder inventory, EDI [Garner, 1984; Paul & Thiel, 2005]). We were able to extend these results in a study on anorectic females by showing that the perception of bodily signals which was assessed in heartbeat perception tests is also decreased in anorexia nervosa (Pollatos et al., 2008). Patients with anorexia nervosa had not only problems in recognizing certain visceral sensations related to hunger and satiety but also exhibited a generally reduced capacity to accurately perceive cardiac bodily signals.

Referring to Fuchs and Schlimme (2009) EDs can be mainly characterized by affection on level of the physical body that one can perceive (object body [Körper]), leading to the assumption of main disturbances of embodiment related to the body image or explicit body awareness. This hypothesis is in accordance to numerous papers reporting disturbances of the body image in EDs (e.g., Rodgers & Chabrol, 2009; Tury, Gülec, & Kohls, 2010). Additionally, there is ample empirical evidence for affected autonomic response patterns and blunted bodily reactions in anorexia nervosa (Murialdo et al., 2007; Zonnevylle-Bender et al., 2005) that in turn favors altered feedback from the body. This fact could constitute an important contributing factor to the onset and maintenance of psychopathology in EDs. The body image is an important part of a person's self-concept, so we want to end up with final ideas concerning the relationship between embodiment, interoception, and the self.

4.3. Implications for our "self"

According to the presented overview, interoceptive states and emotional feelings are directly related and build the fundament of self-awareness and the "self." The representation of bodily signals and the meta-representation of the state of the body in the brain provide a subjective mental image of the "material self" as a feeling or "sentient" entity (see Craig, 2008), that is, the anatomical basis for emotional awareness. Imaging studies on visual self-recognition confirm the relevance of those brain structures that subserve interoceptive integration and emotional awareness, especially the right AIC and the ACC, suggesting that this central network gives rise to an abstract presentation of oneself that could possibly participate in maintaining a sense of self (Devue & Brédart, 2010; Devue et al., 2007).

The embodied fundament of our self has also been highlighted by Damasio (1994, 1999), who suggested a hierarchy of "selves" that are basically constituted by ongoing bodily reactions fed back to the brain and integrated into "somatic markers." The lowest level represents the "proto self" that is an interconnected and temporarily coherent moment-by-moment collection of neural patterns of the current state of the organism, while the "core self" is a second-order entity that maps the state of the proto self and is accessible to conscious experience. The mentioned models summarize that levels of "consciousness" or self-awareness are continually regenerated in a series of pulses of bodily signals which obviously blend together to give rise to a continuous "stream of consciousness." This idea is conceptually connected with earlier views of thinkers belonging to the empiricist tradition like William James. James, who was the first to recognize the importance of bodily reactions

for human feelings and thought, described the unbroken flow of thought, and awareness of the waking mind and consciousness as a stream or a continuous succession of experiences (James, 1890). Accordingly, he proposed that consciousness is based upon basic bodily functions and feedback.

Regarding emotional awareness, it is suggested that in principle emotions are ongoing and continuous behaviors that can also occur *without* subjective feelings (Craig, 2008). This seems to be the case in non-humanoid primates and during non-conscious human states and also, to a certain extent, in specific psychopathological disorders and specific personality traits that are related to deficits in emotional awareness. Being aware of oneself and reflecting upon oneself relates to an individual's self-concept and plays a crucial role in the maintenance of emotional and physical well-being. Being aware of our internal state is relevant for modulating approach and avoidance behaviors that help us to maintain and regain homeostasis, that is, the regulation of internal body state.

Interoception as the fundament of human embodiment offers the theoretical framework with an operationalization of terms and definitions as well as one practical access of research to investigate the embodiment of affective and cognitive processes and self-awareness. It is up to future research to shed more light on what has been called "mind-body-interactions" for human well-being as well as mental and somatic disorders that are all fundamentally related to the self.

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